Priority Messenger System (پیام رسان اولویتی)

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1. Objective:

The purpose of this project is to implement a priority-based messaging system that ensures messages with higher priority are sent first, and messages of the same priority are sent based on their send time.

2. Features:

- Object-oriented design using C++ with classes

- Template-based Queue and derived MaxPriorityQueue

- Message priorities: Text > Image > Audio > Video

- File-based message storage per user

- Time simulation using UNIX timestamps

- Sending speed limitation: 500 bytes/sec

- Statistics: message volume and count between users

3. Implementation:

- **Message Class**: Stores all details of a message (IDs, type, content ID, volume, timestamps, status)

- **Queue Class**: Template-based FIFO queue

- **MaxPriorityQueue Class**: Derived from Queue, dequeues messages based on priority and send time

- **File Handling**: Reads from and writes to user files

- **Simulation**: Calculates send time based on volume and sends messages

- **Statistics**: Prints total bytes and messages per user pair

4. Input/Output:

- Input files are in the format:
DesId|SourceId,MessageType,ContentId,VolumeByte,SendTime|ReceiveTime,Status

- Output: Console messages for sent messages and updated files

5. Challenges:

- Implementing priority logic with templates

- Handling multiple message types and equal priorities

- Simulating time-based sending accurately

6. Conclusion:

This project successfully demonstrates a priority messaging system in C++ with proper OOP design, file I/O handling, and simulation of message sending based on size and priority.